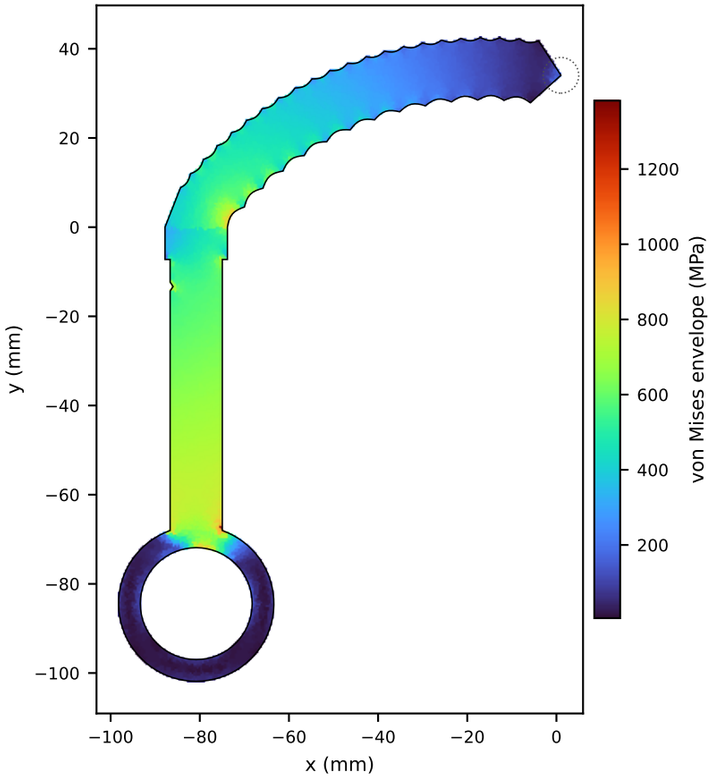
Wave 5 — legal-blade smoke (smoke_legal_blade, first legal winner, post blade-angle knob) | Fitness 7.045 N/g | Est. weight 48.9 g

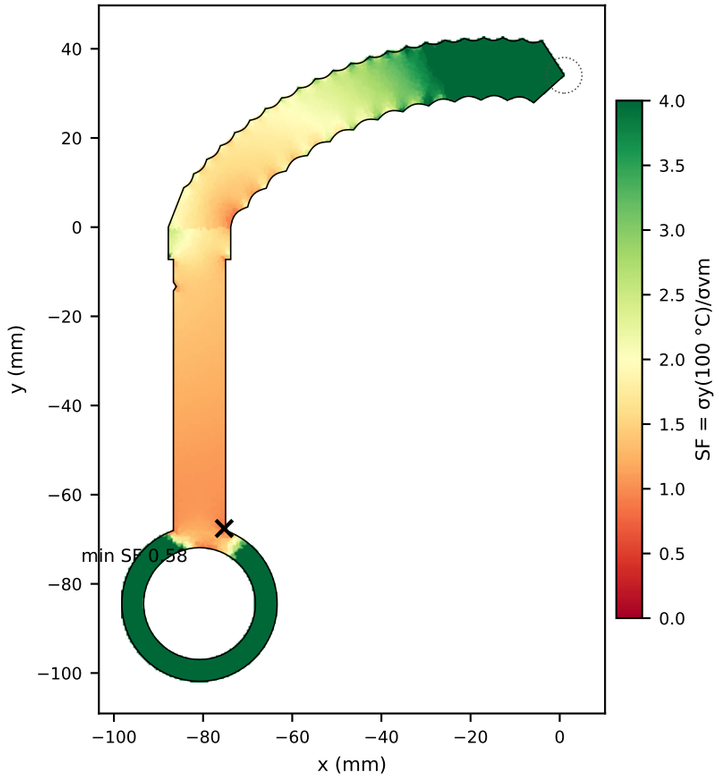
FEM-UNCALIBRATED: offline Kirchhoff plate (numerical DKT + CST, 6657 tris) — not yet calibrated to MC / repo FEM. Envelope of hunter-lateral 260 N, twist 1.26 N·m, tip-pry 260 N, ring-pull 1500 N.

FEM-UNCALIBRATED **blade ≥ 3.375 in: PASS** **hard geometric gates: red-team Ok** **floors 180/260/1500 N on plate: FAIL**

(1) von Mises — worst load case per element



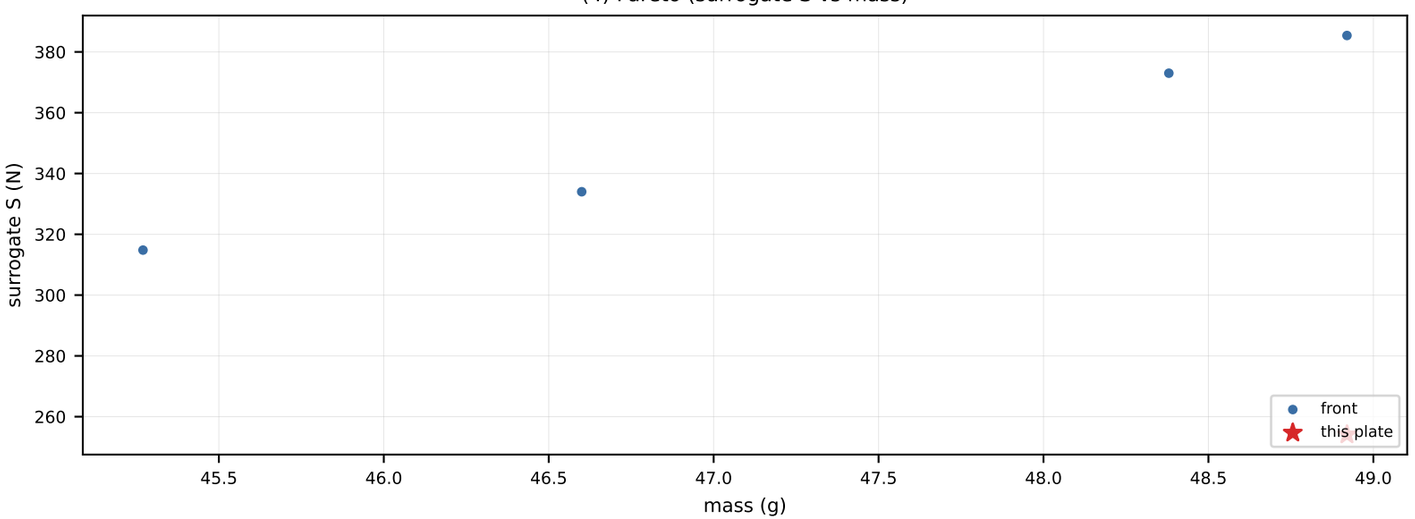
(2) Safety factor — worst case @100 °C



(3) Metrics

Metric	Value	Gate / note
Blade tip→heel	3.526 in (89.6 mm)	**≥3.375 in: PASS**
Blade horizontal reach	3.496 in (88.8 mm)	**≥3.375 in: PASS**
Blade edge arc (outer)	3.964 in (100.7 mm)	**≥3.375 in: PASS**
Grip (ring-neck→guard)	60.8 mm	60–80 mm
Ring ID	25.10 mm	25.4±0.5 mm
Stock / material	4.5 mm / Ti-6Al-4V	red-team: Ok
Est. weight	48.92 g	geom check 48.92 g
Fitness	7.045 N/g	robust plate N/g (FEM-in-loop)
Beam surrogate S	254 N	repo plate (fem.rs): 385 N
Plate FEM @ -10 °C	lat 171 N | ring 6710 N	SF hiker 0.95 / hunter 0.66 / ring 4.47
Plate FEM @ +20 °C	lat 166 N | ring 6495 N	SF hiker 0.92 / hunter 0.64 / ring 4.33
Plate FEM @ +100 °C	lat 151 N | ring 5924 N	SF hiker 0.84 / hunter 0.58 / ring 3.95
Twist / pry SF @100 °C	twist 12.02 | pry 1.95	twist 1.26 N·m, pry 260 N

(4) Pareto (surrogate S vs mass)



Floors on offline plate, worst of -10/20/100 °C: hiker FAIL · hunter FAIL · ring PASS
Temp: Ti yield linear derate (-0.11 %/°C from 20 °C); maps use 100 °C yield 803 MPa.
Serration knock-down on blade (h/h_eff=1.17, from history genes). Load patch r≤4 mm excluded from peaks.

source: out/smoke_legal_blade | tip=farthest blade node (raked) | Jepson Taylor · jarambit (titanium hiking/utility karambit) · generated offline (ops/wave_figures)